

TECHNICAL DESIGN: MULTI-SOURCE CANDIDATE DATA TRANSFORMER

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1. Ingestion & Transformation Pipeline

The transformer operates as a deterministic, multi-stage processing pipeline built on Clean Architecture principles:

- **Ingest & Parse:** Raw byte streams are read by file type (Pandas for recruiter CSV exports; pdfplumber and regex for unstructured PDF resumes) and extracted into intermediate dictionary records.
- **Cleanse & Normalize:** Normalizes phone numbers (E.164 via phonenumbers library), country names/codes (ISO-3166-1 alpha-2 via pycountry), date fields (standardizes 'Jan 2024' or 'Present' to YYYY-MM), and canonicalizes skills.
- **Merge & Deduplicate:** Unifies overlapping candidate records based on source confidence weights (CSV = 0.95, Resume = 0.85, Notes = 0.60). Arrays are unioned, and nested object trees are recursively merged.
- **Score Confidence:** Computes profile completeness based on presence of key profile attributes weighted by the source's trust score.
- **Project to Schema:** Filters, renames paths, and formats the output dynamically at runtime based on the projection configuration.
- **Validate:** Ensures final structural integrity and JSON serializability, degrading gracefully on missing optional fields.

2. Canonical Output Schema & Normalizations

The pipeline resolves messy inputs into a single canonical candidate profile adhering to standard schemas:

- **candidate_id** (string) & **full_name** (string) & **headline** (string or null).
- **emails** (list of lowercase strings) & **phones** (E.164 strings).
- **location:** Structured city, region, and ISO alpha-2 country code.
- **links:** Structure with linkedin, github, portfolio, and other list.
- **skills:** Structured as [{ name, confidence, sources[] }] representing canonical skill names, max confidence, and contributing sources.
- **experience:** Structured as [{ company, title, start, end, summary }] (dates formatted as YYYY-MM).
- **education:** Structured as [{ institution, degree, field, end_year }] (graduation year formatted as YYYY).
- **provenance:** Structured as [{ field, source, method }] to track precisely where each field value came from and the extraction method used.
- **overall_confidence:** Calculated completeness metric based on presence of key profile attributes weighted by the source's trust score.

3. Merge, Conflict Resolution & Trust Policy

Candidate duplicates are resolved across sources by sorting the raw candidate segments in descending order of source confidence. For scalar values (e.g. name, headline, location), the highest-trust source value is selected, and its origin is logged in the provenance tracker. Arrays (emails, phones, other links) are deduplicated and combined. Overlapping experience and education entries are matched by key (e.g., company+title) and non-empty values are merged.

Core Philosophy: 'Wrong-but-confident is worse than honestly-empty.'

To protect downstream hiring systems, empty/null values are preferred over guessed or default-filled records. The Normalizer enforces strict validation: phone numbers that are impossible are set to null; date strings that cannot be formatted are left empty; and unmapped countries are set to null instead of loading dummy placeholders. Invalid data segments are discarded early rather than polluting downstream profiles.

4. Runtime Custom-Output Configuration

The pipeline supports a projection config payload that reshapes output structures without altering code logic:

- **Subsetting & Renaming:** Resolves field selections, including list indicators (emails[0]) and wildcards (skills[].name). Paths are resolved into target keys (e.g., mapping phones[0] to phone).
- **Metadata Toggles:** The config toggle can strip or retain the provenance and overall_confidence sections dynamically.
- **Missing Values:** Controls behavior when selected fields are empty. on_missing enum options are: *null* (retains field with null value), *omit* (removes key from output dictionary), or *error* (raises HTTP 400 validation error).

5. Handled Edge Cases & Scope Limits

- **Malformed/Corrupt Uploads:** Blank files or bad CSV columns are intercepted via try-except guards inside pandas, degrading to empty records.
- **Impossible Phone Formats:** Multi-region format parsing catches invalid length strings, returning None to satisfy the honestly-empty principle.
- **Ambiguous Dates:** Standalone years and sentinel values ('Present', 'Current') are normalized uniformly without crashing calculations.
- **List Wildcard Path Projects:** Resolves list-comprehension paths dynamically (e.g., mapping lists of skill objects to flat string arrays).
- **Deliberately Descope under Time Constraints:** GitHub and LinkedIn live API calls were descope in favor of robust local PDF resume parsing. The ingestion of complex DOCX files was also postponed, with all text parser engines designed around standard PDF structures.